



Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

MTE LAB REPORT – 02

Course No.: **MTE 3202**

Course Title: **Power Electronics and Drives Sessional**

Submitted By:

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Date of Experiment: January 06, 2025

Date of Submission: February 10, 2025

Experiment No.: 02

Name of the Experiment: Observation of A Single-Phase Half Wave Rectifier with R Load.

Objective:

1. To understand the concept of diode-based half-wave rectification.
2. To analyse and observe the waveform of the rectified output graphically.

Theory:

A half-wave rectifier is an electrical circuit designed to transform alternating current (AC) into pulsating direct current (DC) by permitting current flow during only one half of the AC cycle. This is accomplished using a diode, which acts as a one-way switch. During the positive half-cycle of the AC input, the diode is forward-biased, allowing current to pass through, whereas in the negative half-cycle, it becomes reverse-biased and prevents current flow. Consequently, only the positive half of the waveform reaches the output, while the negative half is blocked, producing a pulsating DC output [1].

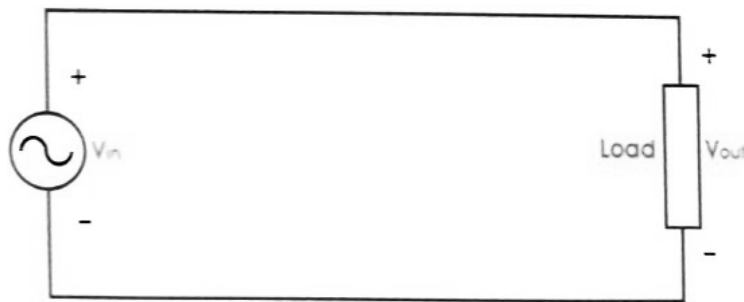


Fig 2.1: Forward Biased Circuit [4]

In a single-phase half-wave rectifier with a resistive (R) load, the output voltage remains directly proportional to the input AC waveform throughout the conducting phase. Several essential parameters help evaluate the performance of a half-wave rectifier. The peak voltage (V_m) denotes the highest voltage attained in the rectified waveform. The DC output voltage (V_{DC}), which represents the average value of the rectified waveform over an entire cycle, is mathematically given as:

$$V_{DC} = \frac{V_m}{\pi}$$

Similarly, the root mean square (RMS) voltage (V_{rms}), which indicates the effective voltage, is expressed as:

$$V_{rms} = \frac{V_m}{2}$$

The ripple factor (RF) quantifies the AC component present in the DC output signal and is determined using the following equation:

$$RF = \sqrt{\frac{V_{rms}^2}{(V_{DC})} - 1}$$



Fig 2.2: Reversed Biased Circuit [4]

Another crucial parameter is the form factor (FF), which is the ratio of V_{rms} to V_{DC} , calculated as:

$$FF = \frac{V_{rms}}{V_{dc}}$$

A half-wave rectifier circuit generally comprises an AC voltage source, a diode, and a resistive load arranged in series. The AC voltage source supplies the input signal, while the diode functions as the rectifier, permitting current flow only during the positive half-cycle. The resistive load consumes rectified power. The voltage measured across the load is a pulsating DC waveform that corresponds to the positive half of the AC signal input. However, due to the elimination of the negative half-cycles, this output experiences significant fluctuations, known as ripples, making it less stable than a full-wave rectifier [2].

This experiment aims to observe the rectified waveform and examine its behavior in a half-wave rectifier with a resistive load. In real-world applications, capacitors or other filtering components are often used to smooth the pulsating DC output. However, this study specifically focuses on analyzing the unfiltered waveform to understand the fundamental rectification process. Half-wave rectifiers are commonly employed in low-power applications, such as signal demodulation circuits and basic power supplies for small DC devices [3].

Required Apparatus:

1. Resistor
2. AC supply
3. Diode
4. Oscilloscope

Circuit Diagram:

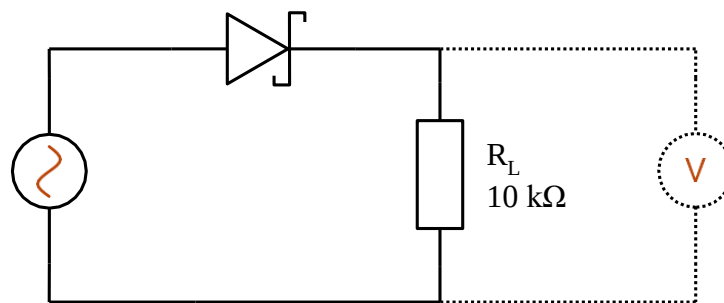


Fig 2.3: Circuit Diagram

Data table and calculation:

Output, V_{pp}	10.20 V
Input, V_{pp}	21.20 V
V_{DC}	$\frac{Vm}{n} = \frac{10.6}{n} = 3.374\text{ V}$
V_{rms}	$\frac{Vm}{2} = \frac{10.6}{2} = 5.3\text{ V}$
Ripple Factor	$\sqrt{\frac{V_{rms}^2}{V_{dc}^2} - 1} = 1.21$
Form Factor (FF)	$\frac{V_{rms}}{V_{dc}} = 1.57$

Oscilloscope Output:

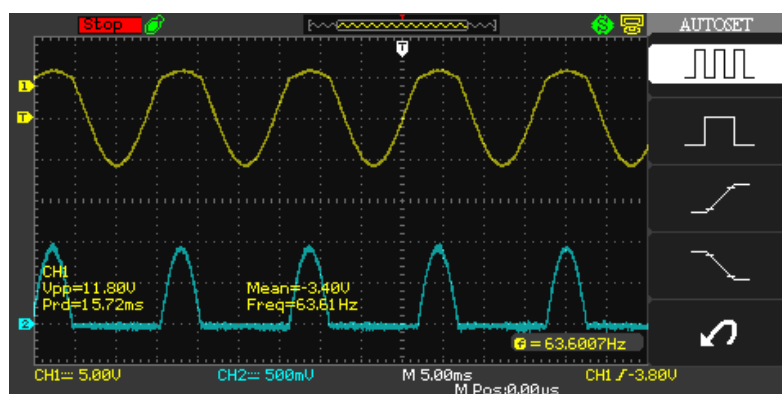


Fig 2.4: Oscilloscope Output of Single-Phase Half Wave Rectifier with R Load

Discussion:

In this experiment, the single-phase half-wave rectifier with a resistive load generated an output peak-to-peak voltage of 10.20V from an input voltage of 21.20V. The calculated DC voltage was 3.374V, while the RMS voltage measured 5.3V. With a ripple factor of 1.21, the output exhibited noticeable fluctuations, which are characteristic of half-wave rectifiers. Additionally, the form factor of 1.5708 indicates a significant difference between the RMS and DC voltages. These findings emphasize the inherent drawbacks of half-wave rectifiers, such as high ripple, low efficiency, and potential power losses due to the inability to produce a smooth DC output.

Conclusion:

The experiment validated the expected behavior of a single-phase half-wave rectifier, demonstrating substantial ripple and a high form factor. Although the output DC voltage was present, its instability due to ripple made the rectification process inefficient. These results underscore the limitations of half-wave rectifiers, particularly in applications that demand a stable and efficient DC power supply.

References:

- [1] R. L. Boylestad and L. Nashelsky, *Electronic Devices and Circuit Theory*, 11th ed., Pearson Education, 2012.
- [2] Ayushi, "Single Phase Half Wave Rectifier- Circuit Diagram, Theory & Applications," *Electrical Volt*, May 25, 2020. <https://www.electricalvolt.com/single-phase-half-wave-rectifier-circuit-diagramtheory-applications/>
- [3] A.S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed., Oxford University Press, 2015.
- [4] "Diode as a Rectifier," Electronics Tutorials, Accessed: Jan. 25, 2025. [Online]. Available: <https://www.electronics-tutorials.ws>